Supplemental Material for Robust Policy Evaluation in Image-Based Contextual Bandits via Vision Transformers

Xuewei Cheng[1,2], John Pleines[2], Shujie Ma[2,*]
[1]*Department of Statistics and Data Science, School of Mathematics and Statistics, Hunan Normal University*, Changsha, China
[2]*Department of Statistics, University of California, Riverside*, Riverside, CA, USA
xwcheng@hunnu.edu.cn, jplei003@ucr.edu, shujie.ma@ucr.edu
[*]Corresponding author

APPENDIX A
NOTATION TABLE

For ease of reading the proofs in the appendix, we provide a summary of the notation in Table III.

TABLE III
NOTATION SUMMARY FOR SAMPLING AND TRANSFORMER ARCHITECTURE.

Notation	Description	Dimension
n	Evaluation sample size	Scalar
m	Nuisance-training sample size	Scalar
N	Total dataset size before splitting	Scalar
$\mathcal{S}_m$	Logged nuisance-training sample	m observations
S_m	Task-specific supervised training sample	m input–label pairs
$X_i^{\mathrm{tr}}, y_i^{\mathrm{tr}}$	Training input and label	Task-dependent
d	Maximum of model, head, and FFN dimensions	Scalar
d_m	Maximum dimension for the class $\mathcal{P}_m$	Scalar
T	Sequence length	Scalar
d_{model}	Model dimension	Scalar
d_h	Head dimension	Scalar
d_f	FFN hidden dimension	Scalar
H	Number of attention heads	Scalar
L	Number of layers	Scalar
C	Number of classes	Scalar
D_0	Token input dimension Td_{model}	Scalar
B_ℓ	Uniform upper bound on the loss	Scalar
p_ζ	Number of fitted head parameters	Scalar
$\mathcal{P}, \mathcal{F}$	Predictor class and scalar loss class	Function classes
$X^{(l)}$	Output of FFN at layer l	$\mathbb{R}^{T\times d_{\mathrm{model}}}$
$Z^{(l)}$	Output of attention at layer l	$\mathbb{R}^{T\times d_{\mathrm{model}}}$
$W_Q^{(l,h)}$	Query weight matrix for head h at layer l	$\mathbb{R}^{d_{\mathrm{model}}\times d_h}$
$W_K^{(l,h)}$	Key weight matrix for head h at layer l	$\mathbb{R}^{d_{\mathrm{model}}\times d_h}$
$W_V^{(l,h)}$	Value weight matrix for head h at layer l	$\mathbb{R}^{d_{\mathrm{model}}\times d_h}$
$W_O^{(l)}$	Output projection matrix at layer l	$\mathbb{R}^{d_{\mathrm{model}}\times d_{\mathrm{model}}}$
$W_1^{(l)}$	Hidden weight matrix of FFN at layer l	$\mathbb{R}^{d_{\mathrm{model}}\times d_f}$
$W_2^{(l)}$	Output weight matrix of FFN at layer l	$\mathbb{R}^{d_f\times d_{\mathrm{model}}}$

Notation. For a positive integer q, define $[q] = \{1, \ldots, q\}$. For a vector $\mathbf{x} = (x_1, \ldots, x_d)^\top$, its L_p-norm is $\|\mathbf{x}\|_p = \left(\sum_{i=1}^d |x_i|^p\right)^{1/p}$. $|\mathcal{J}|$ represents the cardinality of the set $\mathcal{J}$. Let $\|A\|_\sigma$ and $\|A\|_F$ denote the spectral norm (maximum singular value) and the Frobenius norm of A, respectively. We adopt the mathematical notation that $f(x) = O(g(x))$ means $f(x)$ grows at most as fast as $g(x)$. A function $f(x)$ is said to be $o(1)$ if $\lim_{x\to\infty} f(x) = 0$. $o_p(1)$ denotes a sequence of random variables converging to zero in probability, while $O_p(1)$ denotes a sequence that is bounded in probability. More generally, for a positive sequence a_q, $b_q = o_p(a_q)$ iff $b_q/a_q = o_p(1)$, and $b_q = O_p(a_q)$ iff $b_q/a_q = O_p(1)$.

APPENDIX B
TASK-SPECIFIC HEADS FOR REGRESSION AND CLASSIFICATION

We discuss the prediction head used in our architecture. Given the sequence of output representations from the final transformer layer, a pooling operation is first applied to aggregate token-level information into a fixed-size vector. Three common pooling strategies are considered below. The pooled representation is then fed into a task-specific regression or classification head to produce the final prediction.

A. Pooling operations

Common pooling strategies are

$$\begin{aligned}
\text{CLS pooling} \quad & z = X^{(L)}[1] \quad \text{(first [CLS] token)} \\
\text{Mean pooling} \quad & z = \frac{1}{T}\sum_{t=1}^{T} X^{(L)}[t] \\
\text{Max pooling} \quad & z = \max_{t\in\{1,\ldots,T\}} X^{(L)}[t].
\end{aligned}$$

B. Regression head

For regression tasks with target $y \in \mathbb{R}$, the head g_{r} is defined as

$$\hat{y} = g_r(X^{(L)}) = w_r^\top \cdot \mathrm{Pool}(X^{(L)}) + b_r,$$

Here $w_r \in \mathbb{R}^{d_{\mathrm{model}}}$ is the regression weight vector, $b_r \in \mathbb{R}$ is the bias, and $\hat{y} \in \mathbb{R}$ is the predicted value. For reward estimation, an action-specific pair $(w_{r,a}, b_{r,a})$ defines each coordinate $\hat{r}(X, a)$.

C. Classification head

For classification tasks with C classes, the head g_c produces class probabilities

$$\hat{\mathbf{p}} = g_c(X^{(L)}) = \text{Softmax}\left(W_c \cdot \text{Pool}(X^{(L)}) + \mathbf{b}_c\right),$$

Here $W_c \in \mathbb{R}^{C \times d_{\text{model}}}$ is the classification weight matrix, $\mathbf{b}_c \in \mathbb{R}^C$ is the bias, and $\hat{\mathbf{p}} \in \mathbb{R}^C$ satisfies $\sum_{a=1}^{C} \hat{p}_a = 1$. In propensity estimation $C = |\mathcal{A}|$. These head parameters are fixed in Theorems 1 and 2, and may be fitted under Proposition 2.

APPENDIX C
PROOF OF THEOREM 1

Lemma 3 (Lipschitz property of Transformer)**.** *For a common input $\widehat{X}^{(0)} = \widetilde{X}^{(0)} = X^{(0)}$, let $\Delta^{(l)} = \|\widehat{X}^{(l)} - \widetilde{X}^{(l)}\|_F$, so that $\Delta^{(0)} = 0$. Under Assumption 1, for any two parameter configurations $\widehat{W}$ and $\widetilde{W}$ in the bias-free parameterization above, the output difference in Frobenius norm at layer L with H heads satisfies*

$$\begin{aligned}
\Delta^{(L)} &\le \sum_{l=1}^{L} \gamma_2^{L-l} \gamma_1^{(l)} \\
&= \sum_{l=1}^{L} \gamma_2^{L-l} \Bigg[\alpha_1 \max_{h\in[H]} \|\widehat{W}_Q^{(l,h)} - \widetilde{W}_Q^{(l,h)}\|_F \\
&\quad + \alpha_1 \max_{h\in[H]} \|\widehat{W}_K^{(l,h)} - \widetilde{W}_K^{(l,h)}\|_F \\
&\quad + \alpha_2 \max_{h\in[H]} \|\widehat{W}_V^{(l,h)} - \widetilde{W}_V^{(l,h)}\|_F \\
&\quad + \alpha_2 \|\widehat{W}_O^{(l)} - \widetilde{W}_O^{(l)}\|_F + \alpha_2 \|\widehat{W}_1^{(l)} - \widetilde{W}_1^{(l)}\|_F \\
&\quad + \alpha_2 \|\widehat{W}_2^{(l)} - \widetilde{W}_2^{(l)}\|_F \Bigg],
\end{aligned}$$

where $\gamma_1^{(l)} = \alpha_1 \max_{h\in[H]} \|\widehat{W}_Q^{(l,h)} - \widetilde{W}_Q^{(l,h)}\|_F + \alpha_1 \max_{h\in[H]} \|\widehat{W}_K^{(l,h)} - \widetilde{W}_K^{(l,h)}\|_F + \alpha_2 \max_{h\in[H]} \|\widehat{W}_V^{(l,h)} - \widetilde{W}_V^{(l,h)}\|_F + \alpha_2 \|\widehat{W}_O^{(l)} - \widetilde{W}_O^{(l)}\|_F + \alpha_2 \|\widehat{W}_1^{(l)} - \widetilde{W}_1^{(l)}\|_F + \alpha_2 \|\widehat{W}_2^{(l)} - \widetilde{W}_2^{(l)}\|_F$ *and* $\gamma_2 = 2\sqrt{H} B_x^2 B_w^6/\sqrt{d_h} + \sqrt{TH} B_w^4$ *with* $\alpha_1 = \sqrt{H} B_x^3 B_w^5/\sqrt{d_h}$ *and* $\alpha_2 = \sqrt{TH} B_x B_w^3$.

Proof. Step 1. Bounds for query, key, and value matrices.

$$\begin{aligned}
\|Q_h^{(l)}\|_\sigma &= \|X^{(l-1)} W_Q^{(l,h)}\|_\sigma \le B_x B_w \\
\|K_h^{(l)}\|_\sigma &= \|X^{(l-1)} W_K^{(l,h)}\|_\sigma \le B_x B_w \\
\|V_h^{(l)}\|_\sigma &= \|X^{(l-1)} W_V^{(l,h)}\|_\sigma \le B_x B_w
\end{aligned}$$

$$\begin{aligned}
&\|\widehat{Q}_h^{(l)} - \widetilde{Q}_h^{(l)}\|_F = \|\widehat{X}^{(l-1)} \widehat{W}_Q^{(l,h)} - \widetilde{X}^{(l-1)} \widetilde{W}_Q^{(l,h)}\|_F \\
&\quad = \|\widehat{X}^{(l-1)} \widehat{W}_Q^{(l,h)} - \widehat{X}^{(l-1)} \widetilde{W}_Q^{(l,h)} \\
&\qquad + \widehat{X}^{(l-1)} \widetilde{W}_Q^{(l,h)} - \widetilde{X}^{(l-1)} \widetilde{W}_Q^{(l,h)}\|_F \\
&\quad \le \|\widehat{X}^{(l-1)}\|_\sigma \|\widehat{W}_Q^{(l,h)} - \widetilde{W}_Q^{(l,h)}\|_F \\
&\qquad + \|\widetilde{W}_Q^{(l,h)}\|_\sigma \|\widehat{X}^{(l-1)} - \widetilde{X}^{(l-1)}\|_F \\
&\quad \le B_x \|\widehat{W}_Q^{(l,h)} - \widetilde{W}_Q^{(l,h)}\|_F + B_w \|\widehat{X}^{(l-1)} - \widetilde{X}^{(l-1)}\|_F \\
&\|\widehat{K}_h^{(l)} - \widetilde{K}_h^{(l)}\|_F \\
&\quad \le B_x \|\widehat{W}_K^{(l,h)} - \widetilde{W}_K^{(l,h)}\|_F + B_w \|\widehat{X}^{(l-1)} - \widetilde{X}^{(l-1)}\|_F \\
&\|\widehat{V}_h^{(l)} - \widetilde{V}_h^{(l)}\|_F \\
&\quad \le B_x \|\widehat{W}_V^{(l,h)} - \widetilde{W}_V^{(l,h)}\|_F + B_w \|\widehat{X}^{(l-1)} - \widetilde{X}^{(l-1)}\|_F
\end{aligned}$$

Step 2. Bounds for $\text{head}_h^{(l)}$.

Softmax is applied row-wise. Each row of the resulting attention matrix is a probability vector and hence has Euclidean norm at most one. Its Frobenius norm is therefore at most $\sqrt{T}$, which gives

$$\begin{aligned}
\|\text{head}_h^{(l)}\|_\sigma &\le \|\text{head}_h^{(l)}\|_F \\
&\le \left\| \text{softmax}\left(\frac{Q_h^{(l)} (K_h^{(l)})^\top}{\sqrt{d_h}} \right) \right\|_F \|V_h^{(l)}\|_\sigma \\
&\le \sqrt{T} B_x B_w.
\end{aligned}$$

Row-wise softmax is also 1-Lipschitz in Frobenius norm, since its Jacobian on each row is positive semidefinite with spectral norm at most one. Combining this property with the bounds in Step 1 yields

$$\begin{aligned}
&\|\widehat{\text{head}}_h^{(l)} - \widetilde{\text{head}}_h^{(l)}\|_F \\
&\quad \le \|\text{softmax}\left(\frac{\widehat{Q}_h^{(l)} (\widehat{K}_h^{(l)})^\top}{\sqrt{d_h}} \right)\|_\sigma \|\widehat{V}_h^{(l)} - \widetilde{V}_h^{(l)}\|_F \\
&\qquad + \|\widetilde{V}_h^{(l)}\|_\sigma \frac{1}{\sqrt{d_h}} \|\widehat{Q}_h^{(l)} (\widehat{K}_h^{(l)})^\top - \widetilde{Q}_h^{(l)} (\widetilde{K}_h^{(l)})^\top\|_F \\
&\quad \le \sqrt{T} \|\widehat{V}_h^{(l)} - \widetilde{V}_h^{(l)}\|_F + \frac{B_x^2 B_w^2}{\sqrt{d_h}} \|\widehat{Q}_h^{(l)} - \widetilde{Q}_h^{(l)}\|_F \\
&\qquad + \frac{B_x^2 B_w^2}{\sqrt{d_h}} \|\widehat{K}_h^{(l)} - \widetilde{K}_h^{(l)}\|_F \\
&\quad \le \frac{B_x^3 B_w^2}{\sqrt{d_h}} \|\widehat{W}_Q^{(l,h)} - \widetilde{W}_Q^{(l,h)}\|_F \\
&\qquad + \frac{B_x^3 B_w^2}{\sqrt{d_h}} \|\widehat{W}_K^{(l,h)} - \widetilde{W}_K^{(l,h)}\|_F \\
&\qquad + \sqrt{T} B_x \|\widehat{W}_V^{(l,h)} - \widetilde{W}_V^{(l,h)}\|_F \\
&\qquad + \left(2\frac{B_x^2 B_w^3}{\sqrt{d_h}} + \sqrt{T} B_w \right) \|\widehat{X}^{(l-1)} - \widetilde{X}^{(l-1)}\|_F
\end{aligned}$$

Step 3. Bounds for $Z^{(l)}$. Using the Frobenius-norm bound from Step 2 and the identity for the Frobenius norm of a

horizontal concatenation, we obtain

$$\begin{aligned}\|Z^{(l)}\|_\sigma &\le \|Z^{(l)}\|_F \\ &\le \|\mathrm{Concat}\big(\mathrm{head}_1^{(l)},\dots,\mathrm{head}_H^{(l)}\big)\|_F \|W_O^{(l)}\|_\sigma \\ &= \left(\sum_{h=1}^{H}\|\mathrm{head}_h^{(l)}\|_F^2\right)^{1/2}\|W_O^{(l)}\|_\sigma \le \sqrt{TH}B_xB_w^2.\end{aligned}$$

$$\begin{aligned}&\|\widehat{Z}^{(l)}-\widetilde{Z}^{(l)}\|_F \\ &\le \sqrt{TH}B_xB_w\|\widehat{W}_O^{(l)}-\widetilde{W}_O^{(l)}\|_F \\ &\quad + B_w\sqrt{\sum_{h\in[H]}\|\widehat{\mathrm{head}}_h^{(l)}-\widetilde{\mathrm{head}}_h^{(l)}\|_F^2} \\ &\le \frac{\sqrt{H}B_x^3B_w^3}{\sqrt{d_h}}\max_{h\in[H]}\|\widehat{W}_Q^{(l,h)}-\widetilde{W}_Q^{(l,h)}\|_F \\ &\quad + \frac{\sqrt{H}B_x^3B_w^3}{\sqrt{d_h}}\max_{h\in[H]}\|\widehat{W}_K^{(l,h)}-\widetilde{W}_K^{(l,h)}\|_F \\ &\quad + \sqrt{TH}B_xB_w\max_{h\in[H]}\|\widehat{W}_V^{(l,h)}-\widetilde{W}_V^{(l,h)}\|_F \\ &\quad + \sqrt{TH}B_xB_w\|\widehat{W}_O^{(l)}-\widetilde{W}_O^{(l)}\|_F \\ &\quad + \left(2\frac{\sqrt{H}B_x^2B_w^4}{\sqrt{d_h}}+\sqrt{TH}B_w^2\right)\|\widehat{X}^{(l-1)}-\widetilde{X}^{(l-1)}\|_F\end{aligned}$$

Step 4. Bounds for $X^{(l)}$.

In the bias-free parameterization, $X^{(l)} = \mathrm{ReLU}(Z^{(l)}W_1^{(l)})W_2^{(l)}$. Entrywise ReLU is 1-Lipschitz in Frobenius norm. For any two matrices A and B of the same size,

$$\begin{aligned}&\|\mathrm{ReLU}(A)-\mathrm{ReLU}(B)\|_F^2 \\ &\quad = \sum_{i,j}\big|\max\{A_{ij},0\}-\max\{B_{ij},0\}\big|^2 \le \|A-B\|_F^2.\end{aligned}$$

Taking $B=0$ gives the valid bound

$$\|\mathrm{ReLU}(A)\|_\sigma \le \|\mathrm{ReLU}(A)\|_F \le \|A\|_F.$$

Combining this with the Frobenius-norm bound for $Z^{(l)}$ in Step 3 gives

$$\begin{aligned}\|X^{(l)}\|_\sigma \le \|X^{(l)}\|_F &\le \|\mathrm{ReLU}(Z^{(l)}W_1^{(l)})\|_F\|W_2^{(l)}\|_\sigma \\ &\le \|Z^{(l)}\|_F\|W_1^{(l)}\|_\sigma\|W_2^{(l)}\|_\sigma \le \sqrt{TH}B_xB_w^4.\end{aligned}$$

Consequently, $\|h_W(X^{(0)})\|_F \le B_h = \sqrt{THd}B_xB_w^4$ with $d=\max\{d_h,d_{\mathrm{model}},d_f\}\ge 1$, as used in the subsequent bounds.

For the perturbation bound, add and subtract $\mathrm{ReLU}(\widetilde{Z}^{(l)}\widetilde{W}_1^{(l)})\widehat{W}_2^{(l)}$. The preceding inequalities give

$$\begin{aligned}\Delta^{(l)} &\le \|\mathrm{ReLU}(\widehat{Z}^{(l)}\widehat{W}_1^{(l)})-\mathrm{ReLU}(\widetilde{Z}^{(l)}\widetilde{W}_1^{(l)})\|_F\|\widehat{W}_2^{(l)}\|_\sigma \\ &\quad + \|\mathrm{ReLU}(\widetilde{Z}^{(l)}\widetilde{W}_1^{(l)})\|_\sigma\|\widehat{W}_2^{(l)}-\widetilde{W}_2^{(l)}\|_F \\ &\le \|\widehat{W}_1^{(l)}\|_\sigma\|\widehat{W}_2^{(l)}\|_\sigma\|\widehat{Z}^{(l)}-\widetilde{Z}^{(l)}\|_F \\ &\quad + \|\widetilde{Z}^{(l)}\|_\sigma\|\widehat{W}_2^{(l)}\|_\sigma\|\widehat{W}_1^{(l)}-\widetilde{W}_1^{(l)}\|_F \\ &\quad + \|\widetilde{Z}^{(l)}\|_F\|\widetilde{W}_1^{(l)}\|_\sigma\|\widehat{W}_2^{(l)}-\widetilde{W}_2^{(l)}\|_F \\ &\le B_w^2\|\widehat{Z}^{(l)}-\widetilde{Z}^{(l)}\|_F \\ &\quad + \sqrt{TH}B_xB_w^3\Big(\|\widehat{W}_1^{(l)}-\widetilde{W}_1^{(l)}\|_F \\ &\qquad + \|\widehat{W}_2^{(l)}-\widetilde{W}_2^{(l)}\|_F\Big).\end{aligned}$$

Here the coefficient of $\|\widehat{W}_2^{(l)}-\widetilde{W}_2^{(l)}\|_F$ is controlled through the Frobenius-norm bound for $\widetilde{Z}^{(l)}$ in Step 3. Substituting the perturbation bound from that step yields

$$\begin{aligned}\Delta^{(l)} &\le \frac{\sqrt{H}B_x^3B_w^5}{\sqrt{d_h}}\max_{h\in[H]}\|\widehat{W}_Q^{(l,h)}-\widetilde{W}_Q^{(l,h)}\|_F \\ &\quad + \frac{\sqrt{H}B_x^3B_w^5}{\sqrt{d_h}}\max_{h\in[H]}\|\widehat{W}_K^{(l,h)}-\widetilde{W}_K^{(l,h)}\|_F \\ &\quad + \sqrt{TH}B_xB_w^3\max_{h\in[H]}\|\widehat{W}_V^{(l,h)}-\widetilde{W}_V^{(l,h)}\|_F \\ &\quad + \sqrt{TH}B_xB_w^3\|\widehat{W}_O^{(l)}-\widetilde{W}_O^{(l)}\|_F \\ &\quad + \sqrt{TH}B_xB_w^3\|\widehat{W}_1^{(l)}-\widetilde{W}_1^{(l)}\|_F \\ &\quad + \sqrt{TH}B_xB_w^3\|\widehat{W}_2^{(l)}-\widetilde{W}_2^{(l)}\|_F \\ &\quad + \left(2\frac{\sqrt{H}B_x^2B_w^6}{\sqrt{d_h}}+\sqrt{TH}B_w^4\right)\|\widehat{X}^{(l-1)}-\widetilde{X}^{(l-1)}\|_F \\ &= \gamma_1^{(l)}+\gamma_2\Delta^{(l-1)}.\end{aligned}$$

Iterating this recursion and using $\Delta^{(0)}=0$, we conclude that

$$\Delta^{(L)} \le \sum_{l=1}^{L}\gamma_2^{L-l}\gamma_1^{(l)} + \gamma_2^L\Delta^{(0)} = \sum_{l=1}^{L}\gamma_2^{L-l}\gamma_1^{(l)}.$$

□

Lemma 4 (Covering number bound)**.** *Under Assumption 1, define* $\|h-\tilde{h}\|_{S_m,2}^2 = m^{-1}\sum_{i=1}^m\|h(X_i^{\mathrm{tr}})-\tilde{h}(X_i^{\mathrm{tr}})\|_F^2$. *For every* $\epsilon>0$*, the encoder class satisfies*

$$\log\mathcal{N}(\mathcal{H},\epsilon,\|\cdot\|_{S_m,2}) \le 6Ld^2\log\left(1+\frac{12\gamma HL\sqrt{d}B_w}{\epsilon}\right),$$

where $\mathcal{N}$ *denotes covering numbers with centers in the class, and* d,γ *are as in Theorem 1.*

Proof. Lemma 3 gives $\|h_{\widehat{W}}-h_{\widetilde{W}}\|_{S_m,2} \le (3H+3)L\gamma\max_{j\in\mathcal{J}}\|\widehat{W}_j-\widetilde{W}_j\|_F$. Each matrix lies in a Frobenius ball of radius $\sqrt{d}B_w$. The total number of scalar parameters is

$$\begin{aligned}p_W &= L(3Hd_{\mathrm{model}}d_h + d_{\mathrm{model}}^2 + 2d_{\mathrm{model}}d_f) \\ &= L(4d_{\mathrm{model}}^2+2d_{\mathrm{model}}d_f) \le 6Ld^2.\end{aligned}$$

Take a maximal τ-separated subset of the admissible parameter set in the metric $\max_j \|W_j - \widetilde{W}_j\|_F$. Product balls of radius $\tau/2$ about these points are disjoint and lie in the product of balls of radius $\sqrt{d}B_w + \tau/2$. Comparing volumes bounds the subset size by $(1+2\sqrt{d}B_w/\tau)^{p_W}$. Maximality makes it a τ-net with admissible centers, including when the intermediate-state constraint couples the matrices. Taking $\tau = \epsilon/[(3H+3)L\gamma]$ and using $3H+3 \le 6H$ proves the result. □

Lemma 5 (Rademacher complexity bound)**.** *Let $\mathcal{H}$ be the Transformer encoder class with $0 \in \mathcal{H}$ and $\|h_W(X^{(0)})\|_F \le B_h$, where $B_h = \sqrt{THd}B_x B_w^4$. Under Assumptions 1 and 2, for a fixed $g \in \mathcal{G}$ and any sample $S_m = \{(X_i^{\mathrm{tr}}, y_i^{\mathrm{tr}})\}_{i=1}^m$, the loss-composed class $\mathcal{F} = \{\ell \circ g \circ h_W : h_W \in \mathcal{H}\}$ satisfies*

$$\begin{aligned}\widehat{\mathfrak{R}}_m(\mathcal{F}) &\le \frac{12\sqrt{6}\rho_g\rho_\ell d\gamma B_w\sqrt{L}}{\sqrt{m}}\left[\sqrt{\log(1+12HL\sqrt{d})}+\frac{\sqrt{\pi}}{2}\right]\\ &= O\left(\frac{\rho_g\rho_\ell d\gamma B_w \varsigma\sqrt{L}}{\sqrt{m}}\right),\end{aligned}$$

where $\gamma = \max\{\alpha_1\gamma_2^{L-l}, \alpha_2\gamma_2^{L-l}\}_{l\in[L]}$ and $d = \max\{d_h, d_{\mathrm{model}}, d_f\}$, and $\varsigma = \sqrt{\log(1+12HL\sqrt{d})} + \frac{\sqrt{\pi}}{2}$. The implicit constant is universal and independent of m, d, H, L and the norm and Lipschitz bounds.

Proof. For a class $\mathcal{C}$ of scalar functions on (X, y), define its empirical Rademacher complexity by

$$\widehat{\mathfrak{R}}_m(\mathcal{C}) = \mathbb{E}_\varepsilon\left[\sup_{u\in\mathcal{C}}\frac{1}{m}\sum_{i=1}^m \varepsilon_i u(X_i^{\mathrm{tr}}, y_i^{\mathrm{tr}})\right],$$

where $\varepsilon_1, \dots, \varepsilon_m$ are independent random signs with $\mathbb{P}(\varepsilon_i = 1) = \mathbb{P}(\varepsilon_i = -1) = 1/2$, and the expectation is conditional on S_m. Use the normalized empirical Frobenius distance on encoder outputs and the corresponding scalar norm

$$\|h-\tilde{h}\|_{S_m,2}^2 = \frac{1}{m}\sum_{i=1}^m \|h(X_i^{\mathrm{tr}}) - \tilde{h}(X_i^{\mathrm{tr}})\|_F^2,$$
$$\|u\|_{S_m,2}^2 = \frac{1}{m}\sum_{i=1}^m |u(X_i^{\mathrm{tr}}, y_i^{\mathrm{tr}})|^2,$$

where $h, \tilde{h} \in \mathcal{H}$ and u is scalar-valued. In either case, the unnormalized version is $\|\cdot\|_{S_m,F} = \sqrt{m}\|\cdot\|_{S_m,2}$. Write $\mathcal{N}(\mathcal{C}, \epsilon, \|\cdot\|)$ for the minimum size of an ϵ-cover of a function class $\mathcal{C}$ in the indicated distance, with centers in that class. Then

$$\mathcal{N}(\mathcal{H}, \epsilon, \|\cdot\|_{S_m,F}) = \mathcal{N}(\mathcal{H}, \epsilon/\sqrt{m}, \|\cdot\|_{S_m,2}).$$

Lemma 4 supplies the encoder covering bound in the normalized empirical distance. We transfer this bound directly to the scalar loss class.

Center the loss class at the zero encoder by setting $\mathcal{L}_0 = \{(X, y) \mapsto \ell(g(h_W(X)), y) - \ell(g(0), y) : h_W \in \mathcal{H}\}$. Then $0 \in \mathcal{L}_0$ and $\widehat{\mathfrak{R}}_m(\mathcal{F}) = \widehat{\mathfrak{R}}_m(\mathcal{L}_0)$, since translation by a fixed function leaves empirical Rademacher complexity unchanged. Step 4 of Lemma 3 gives $\|h_W(X)\|_F \le \alpha_2 B_w \le \gamma B_w$. Hence, with $R = \rho_g\rho_\ell\gamma B_w$, every $u \in \mathcal{L}_0$ satisfies $\|u\|_{S_m,2} \le R$, and the Lipschitz properties give

$$\log\mathcal{N}(\mathcal{L}_0, \epsilon, \|\cdot\|_{S_m,2}) \le 6Ld^2\log\left(1+\frac{12HL\sqrt{d}R}{\epsilon}\right).$$

Dudley's entropy integral in the normalized distance [1] now yields

$$\begin{aligned}\widehat{\mathfrak{R}}_m(\mathcal{F}) &\le \frac{12}{\sqrt{m}}\int_0^R \sqrt{\log\mathcal{N}(\mathcal{L}_0, \epsilon, \|\cdot\|_{S_m,2})}\,d\epsilon\\ &\le \frac{12\sqrt{6}d\sqrt{L}R}{\sqrt{m}}\int_0^1 \sqrt{\log(1+12HL\sqrt{d}/t)}\,dt\\ &\le \frac{12\sqrt{6}d\sqrt{L}R}{\sqrt{m}}\left[\sqrt{\log(1+12HL\sqrt{d})}+\frac{\sqrt{\pi}}{2}\right].\end{aligned}$$

The second line uses the change of variables $t = \epsilon/R$. The last line follows from $\log(1+a/t) \le \log(1+a)+\log(1/t)$ for $a > 0$ and $0 < t \le 1$, applied with $a = 12HL\sqrt{d}$, together with $\int_0^1 \sqrt{\log(1/t)}\,dt = \sqrt{\pi}/2$. The entropy integral is finite at zero, which justifies taking the truncation parameter to zero. Since $H, L, d \ge 1$, we have $\log(1+12HL\sqrt{d}) \ge \log 13$. Consequently, the additive constant $\sqrt{\pi}/2$ is absorbed into the stated order with a universal constant. Substituting R proves the lemma. Combining it with the standard bounded-loss Rademacher generalization inequality [2] proves Theorem 1. □

Appendix D
Proof of Theorem 2

Lemma 6 (Local Rademacher complexity)**.** *Under Assumption 4, the centered local loss class $\mathcal{L}_\rho$ satisfies, for every $\rho > 0$,*

$$\mathbb{E}_{S_m}\widehat{\mathfrak{R}}_m(\mathcal{L}_\rho) \le \frac{\rho_\ell\eta\rho}{\sqrt{m}}. \tag{17}$$

Here η is the local entropy coefficient in (4)*. The empirical Rademacher complexity and norm are defined as in Lemma 5.*

Proof. Since $f^* \in \rho\mathcal{B}$, the scalar class $\mathcal{L}_\rho$ contains zero. Applying Dudley's entropy inequality conditionally on S_m, then taking expectations and using (4), gives

$$\begin{aligned}\mathbb{E}_{S_m}\widehat{\mathfrak{R}}_m(\mathcal{L}_\rho) &\le \frac{12}{\sqrt{m}}\mathbb{E}_{S_m}\int_0^\infty \sqrt{\log\mathcal{N}(\mathcal{L}_\rho, \epsilon, \|\cdot\|_{S_m,2})}\,d\epsilon\\ &\le \frac{\rho_\ell\eta\rho}{\sqrt{m}}.\end{aligned}$$

The entropy condition ensures integrability at zero. Extending the upper integration limit to infinity only increases the bound. □

Remark 7. *The coefficient η is constrained by Assumption 4. The loss Lipschitz factor is already included in* (17)*.*

Proof of Theorem 2. Define the event

$$\Omega_0 = \big\{|\mathcal{E}(f) - \widehat{\mathcal{E}}(f)| \le \rho_\ell\theta\max\{\|f-f^*\|_2^2, \varphi^2\} \text{ for all } f \in \mathcal{P}\big\}. \tag{18}$$

On Ω_0, if $\|\hat{f} - f^*\|_2 > \varphi$, Assumption 3 implies

$$\widehat{\mathcal{E}}(\hat{f}) \geq (1/A - \rho_\ell \theta)\|\hat{f} - f^*\|_2^2 > 0,$$

contradicting empirical risk minimization. Hence $\|\hat{f} - f^*\|_2 \leq \varphi$, and

$$\mathcal{E}(\hat{f}) \leq \widehat{\mathcal{E}}(\hat{f}) + \rho_\ell \theta \varphi^2 \leq \rho_\ell \theta \varphi^2.$$

To control the probability of Ω_0, use an inner ball and outer shells

$$\mathcal{P}_1 = \{f \in \mathcal{P} : \|f - f^*\|_2 \leq 2\varphi\},$$
$$\mathcal{P}_k = \{f \in \mathcal{P} : 2^{k-1}\varphi < \|f - f^*\|_2 \leq 2^k \varphi\}, \qquad k \geq 2.$$

These sets cover $\mathcal{P}$, including the ball $\varphi\mathcal{B}$. For each nonempty $\mathcal{P}_k$, define

$$\mathcal{L}_f(X, y) = \ell(f(X), y) - \ell(f^*(X), y),$$
$$Z_k = \sup_{f \in \mathcal{P}_k} |\mathcal{E}(f) - \widehat{\mathcal{E}}(f)|,$$
$$\|\mathcal{L}_{\mathcal{P}_k}\|_\infty = \sup_{f \in \mathcal{P}_k} \|\mathcal{L}_f\|_\infty.$$

The bounded empirical-process inequality in the supplementary proof of Theorem 9.2 of [3] gives, for every $\xi > 0$, with probability at least $1 - 2e^{-\xi}$,

$$Z_k \leq 2\mathbb{E}Z_k + \sup_{f \in \mathcal{P}_k} \sqrt{\mathbb{E}\mathcal{L}_f^2}\sqrt{\frac{8\xi}{m}} + \frac{69\|\mathcal{L}_{\mathcal{P}_k}\|_\infty \xi}{2m}. \tag{19}$$

The Lipschitz and encoder envelope bounds imply

$$\sup_{f \in \mathcal{P}_k} \sqrt{\mathbb{E}\mathcal{L}_f^2} \leq 2^k \rho_\ell \varphi, \qquad \|\mathcal{L}_{\mathcal{P}_k}\|_\infty \leq 2\rho_\ell \rho_g B_h.$$

Symmetrization, followed by bounding an absolute supremum by the sum of the positive and negative suprema over the zero-containing class $\mathcal{L}_{2^k\varphi}$, yields

$$\mathbb{E}Z_k \leq 4\mathbb{E}_{S_m}\widehat{\mathfrak{R}}_m(\mathcal{L}_{2^k\varphi}) \leq \frac{4\rho_\ell \eta\, 2^k \varphi}{\sqrt{m}} \leq \frac{\rho_\ell \theta}{12}\, 2^k \varphi^2.$$

The second inequality uses Lemma 6, and the last uses $\varphi \geq 48\eta/(\sqrt{m}\theta)$.

Set $\xi = m\omega\theta^2(2^{k-1}\varphi)^2$ in (19). Since $2^{k-1} \leq 4^{k-1}$ for $k \geq 1$, with failure probability at most $2e^{-\vartheta 4^{k-1}}$,

$$\begin{aligned} Z_k &\leq \frac{\rho_\ell \theta}{3}\, 2^{k-1}\varphi^2 + 4\sqrt{2\omega}\rho_\ell \theta (2^{k-1}\varphi)^2 \\ &\quad + 69\rho_\ell \omega \rho_g B_h \theta^2 (2^{k-1}\varphi)^2 \\ &\leq \rho_\ell \theta (2^{k-1}\varphi)^2 \left[\frac{1}{3} + 4\sqrt{2\omega} + 69\omega\rho_g B_h \theta\right] \\ &\leq \rho_\ell \theta (2^{k-1}\varphi)^2. \end{aligned}$$

The bracket is at most one by the definition of ω. For $k = 1$ this controls the ball of radius 2φ, including the inner ball, by $\rho_\ell\theta\varphi^2$. For $k \geq 2$, $(2^{k-1}\varphi)^2 < \|f - f^*\|_2^2$ on $\mathcal{P}_k$. Thus the simultaneous events imply Ω_0 over the entire class. A union bound and $4^{k-1} \geq 3k - 2$ give

$$\mathbb{P}(\Omega_0^c) \leq 2\sum_{k=1}^{\infty} e^{-\vartheta 4^{k-1}} \leq \frac{2e^{-\vartheta}}{1 - e^{-3\vartheta}}.$$

For (6), $\vartheta \geq \log(4/\delta)$, and the last expression is at most δ. This proves both claims. □

APPENDIX E
FROZEN-ENCODER NUISANCE ESTIMATION

Proof of Proposition 2. A Euclidean packing argument gives a τ-cover of $\mathcal{B}_\zeta$ of size at most $(1 + 2B_\zeta/\tau)^{p_\zeta}$ with centers in the class. The parameter and loss Lipschitz bounds therefore imply

$$\log \mathcal{N}(\mathcal{F}_{\mathrm{pre}}, \epsilon, \|\cdot\|_{S_m,2}) \leq p_\zeta \log\left(1 + \frac{2\rho_\ell \Lambda_\zeta B_\zeta}{\epsilon}\right),$$

where $\mathcal{F}_{\mathrm{pre}}$ is the scalar loss class. Center this class at any fixed member. Its radius is at most B_ℓ in the empirical norm because both losses lie in $[0, B_\ell]$. The entropy integral argument in Lemma 5 gives

$$\widehat{\mathfrak{R}}_m(\mathcal{F}_{\mathrm{pre}}) \leq 12 B_\ell \sqrt{\frac{p_\zeta}{m}}\left[\sqrt{\log\left(1 + \frac{2\rho_\ell \Lambda_\zeta B_\zeta}{B_\ell}\right)} + \frac{\sqrt{\pi}}{2}\right].$$

Apply the bounded-loss generalization inequality to both ℓ and $B_\ell - \ell$, each with failure probability $\delta/2$. Their empirical Rademacher complexities agree by symmetry of the random signs. This yields the two-sided bound $\Gamma_m(\delta)$. On that event, for every comparator $f \in \mathcal{P}_{\mathrm{pre}}$,

$$\begin{aligned} \mathcal{R}(\hat{f}) &\leq \widehat{\mathcal{R}}(\hat{f}) + \Gamma_m(\delta) \leq \widehat{\mathcal{R}}(f) + \epsilon_{\mathrm{opt}} + \Gamma_m(\delta) \\ &\leq \mathcal{R}(f) + 2\Gamma_m(\delta) + \epsilon_{\mathrm{opt}}. \end{aligned}$$

Taking the infimum proves the risk bound. The squared-loss identity follows by conditioning on (X, a), and the cross-entropy identity follows by conditioning on X. Pinsker's inequality gives the probability-vector bound. □

For an explicit head parameterization, let $z = \mathrm{Pool}(h_{\mathrm{pre}}(X)) \in \mathbb{R}^{d_z}$ with $\|z\|_2 \leq B_z$. Consider

$$g_\zeta(z) = \chi\{V\sigma(Uz + b) + v\},$$

where ζ stacks the entries of (U, b, V, v), the hidden width is q, the output dimension is d_{out}, $\sigma(0) = 0$, and σ is L_σ-Lipschitz. Suppose χ is 1-Lipschitz and each of $\|U\|_F, \|V\|_F, \|b\|_2, \|v\|_2$ is bounded by $B > 0$. Then $p_\zeta = qd_z + q + d_{\mathrm{out}}q + d_{\mathrm{out}}$, $B_\zeta = 2B$, and the triangle inequality gives

$$\begin{aligned} &\|g_\zeta(z) - g_{\zeta'}(z)\|_2 \\ &\quad \leq BL_\sigma B_z \|U - U'\|_F + BL_\sigma \|b - b'\|_2 \\ &\qquad + L_\sigma B(B_z + 1)\|V - V'\|_F + \|v - v'\|_2 \\ &\quad \leq \{2BL_\sigma(B_z + 1) + 1\}\|\zeta - \zeta'\|_2. \end{aligned}$$

Thus $\Lambda_\zeta = 2BL_\sigma(B_z + 1) + 1$ is valid. ReLU and GELU have finite Lipschitz constants and vanish at zero. Softmax, Euclidean projection onto a truncated simplex, scalar clipping, and action-coordinate selection are all nonexpansive in Euclidean norm. For a linear head, the same argument gives $\Lambda_\zeta = \sqrt{B_z^2 + 1}$ under the joint Euclidean norm on its matrix and bias parameters. These examples specify the norm constraints needed for the fitted-head analysis. If the encoder was pretrained on an independent sample, the argument holds conditionally on that sample.

Proof of Corollary 1. Compactness of the coefficient sets and boundedness of z give a constant $c > 0$ such that every propensity coordinate is at least c throughout the class. In particular, the true behavior probabilities satisfy this bound. Correct specification gives Assumption 3 with $A = 1$ for squared loss and $A = 2$ for cross-entropy.

We verify local entropy in the prediction norm of each task. For reward heads,

$$\|r_\zeta - r_{\zeta'}\|_{2,\mu}^2 = \sum_a \mathbb{E}_X[\mu(a \mid X)\{(\zeta_{r,a} - \zeta'_{r,a})^\top z(X)\}^2] \geq c\lambda_z \|\zeta_r - \zeta'_r\|_2^2.$$

For propensity heads, let $s \in \mathbb{R}^{C-1}$ be the free logits and let p_- contain the first $C-1$ probabilities. The Hessian of $\log(1 + \sum_{a<C} e^{s_a})$ is

$$H(s) = \operatorname{diag}(p_-) - p_- p_-^\top \succeq c^2 I_{C-1}.$$

Indeed, for any $v \in \mathbb{R}^{C-1}$, weighted Cauchy–Schwarz gives $v^\top H(s) v \geq p_C \sum_{a<C} p_a v_a^2 \geq c^2 \|v\|_2^2$. Integrating the Hessian along the segment between two admissible coefficient vectors yields $\|p_-(s) - p_-(s')\|_2 \geq c^2 \|s - s'\|_2$. Consequently,

$$\mathbb{E}_X \|\mu_\zeta(\cdot \mid X) - \mu_{\zeta'}(\cdot \mid X)\|_2^2 \geq c^4 \lambda_z \|\zeta_\mu - \zeta'_\mu\|_2^2.$$

Write $\kappa_r = c\lambda_z$ and $\kappa_\mu = c^4 \lambda_z$. A prediction ball of radius ρ is contained in a coefficient ball of radius $\rho/\sqrt{\kappa_j}$ about the true coefficient vector. Both prediction maps are pointwise B_z-Lipschitz in their coefficients. For the task-specific loss Lipschitz constant ρ_ℓ, set $R_\rho = \rho_\ell B_z \rho/\sqrt{\kappa_j}$. A packing argument with centers in the local class gives, for every sample,

$$\log \mathcal{N}(\mathcal{L}_\rho, \epsilon, \|\cdot\|_{S_m,2}) \leq p_j \log(1 + 2R_\rho/\epsilon).$$

The zero function covers $\mathcal{L}_\rho$ at radius R_ρ. Integrating as in Lemma 5 verifies Assumption 4 with

$$\eta_j = \frac{12 B_z \sqrt{p_j}}{\sqrt{\kappa_j}} \left(\sqrt{\log 3} + \frac{\sqrt{\pi}}{2} \right).$$

For the optimization-error bound, let $\widehat{\mathcal{R}}_j(\zeta_j)$ be the empirical loss. The empirical feature second moment converges to $\mathbb{E}_X[zz^\top]$, and each observed-action second moment converges to $\mathbb{E}_X[\mu(a \mid X) zz^\top] \succeq c\lambda_z I_{d_z}$. The preceding Hessian inequality and the squared-loss Hessian therefore give, with probability tending to one, a constant $\underline{\kappa}_j > 0$ such that

$$\widehat{\mathcal{R}}_j(\zeta_j) - \widehat{\mathcal{R}}_j(\zeta_j^0) \geq \nabla \widehat{\mathcal{R}}_j(\zeta_j^0)^\top (\zeta_j - \zeta_j^0) + \frac{\underline{\kappa}_j}{2} \|\zeta_j - \zeta_j^0\|_2^2$$

uniformly over the compact convex parameter set, where ζ_j^0 represents the true nuisance. The gradient at ζ_j^0 is an average of independent, centered, bounded vectors of fixed dimension, so its norm is $O_p(m^{-1/2})$. Approximate empirical minimization gives

$$\frac{\underline{\kappa}_j}{2} \|\hat{\zeta}_j - \zeta_j^0\|_2^2 \leq \|\nabla \widehat{\mathcal{R}}_j(\zeta_j^0)\|_2 \|\hat{\zeta}_j - \zeta_j^0\|_2 + \epsilon_{j,m}.$$

Solving this quadratic inequality yields $\|\hat{\zeta}_j - \zeta_j^0\|_2 = O_p(m^{-1/2} + \sqrt{\epsilon_{j,m}})$. Prediction Lipschitz continuity and $\|u\|_{2,\nu} \leq c^{-1/2} \|u\|_{2,\mu}$ give (11). If $\epsilon_{j,m} = O_p(m^{-1})$, the nuisance-error product is $O_p(m^{-1}) = o_p(n^{-1/2})$ when $m \asymp n$. □

APPENDIX F
DR REMAINDER AND STOCHASTIC-POLICY INFERENCE

Proof of Lemma 1. Pointwise, $|1 - \varrho_k| = |\hat{\mu} - \mu|/\hat{\mu} \leq c^{-1} |\hat{\mu} - \mu|$. Taking the target-weighted L_2 norm and multiplying by $\|\Delta\|_{2,\nu}$ proves (9). The transfer inequality follows from $\nu(a \mid X) \leq 1 \leq \mu(a \mid X)/c$.

If the conditional Hölder rate assumptions stated after the lemma hold, then $\beta_j > D_0/2$ is equivalent to $\beta_j/(2\beta_j + D_0) > 1/4$. Consequently, $m^{-\beta_j/(2\beta_j + D_0)} \log m = o(m^{-1/4})$, and $\eta_{j,m}/\sqrt{m} = o(m^{-1/4})$. Thus both individual errors are $o_p(n^{-1/4})$ when $m \asymp n$, yielding the required product rate. □

Proof of Lemma 2. Write $Z = (X, a, r)$, let P_μ be its stationary logging law, and use $P_\mu u = \mathbb{E}_{P_\mu} u(Z)$ and $P_n u = n^{-1} \sum_{k=1}^n u(Z_k)$ for population and evaluation averages. Conditional on the independent training sample $\mathcal{S}_m$, the fitted score is

$$\hat{\psi}_\nu(Z) = \sum_b \nu(b \mid X) \hat{r}(X, b) + \frac{\nu(a \mid X)}{\hat{\mu}(a \mid X)} \{r - \hat{r}(X, a)\},$$

where b indexes actions. Taking its conditional expectation under the logging law gives the exact bias identity

$$P_\mu \hat{\psi}_\nu - V = \mathbb{E}_X \sum_a \nu(a \mid X) \left\{ 1 - \frac{\mu(a \mid X)}{\hat{\mu}(a \mid X)} \right\} \Delta(X, a),$$

$$|P_\mu \hat{\psi}_\nu - V| \leq \|1 - \varrho_k\|_{2,\nu} \|\Delta\|_{2,\nu} = o_p(n^{-1/2}).$$

The identity establishes conditional unbiasedness when either fitted nuisance equals its true function. In general, conditional bias depends on the joint error product.

The fitted and oracle scores differ by

$$\begin{aligned} \hat{\psi}_\nu(Z) - \psi_\nu(Z) &= \sum_b \nu(b \mid X) \Delta(X, b) - \frac{\nu(a \mid X)}{\hat{\mu}(a \mid X)} \Delta(X, a) \\ &\quad + \nu(a \mid X) \left\{ \frac{1}{\hat{\mu}(a \mid X)} - \frac{1}{\mu(a \mid X)} \right\} \{r - r^*(X, a)\}. \end{aligned}$$

Since $|r - r^*(X, a)| \leq 2M$, Jensen's inequality and both propensity lower bounds imply

$$\begin{aligned} \|\hat{\psi}_\nu - \psi_\nu\|_{L_2(P_\mu)} &\leq (1 + c^{-1}) \|\Delta\|_{2,\nu} + 2Mc^{-2} \|\hat{\mu} - \mu\|_{2,\nu} \\ &= o_p(1). \end{aligned}$$

Conditional independence of the evaluation observations therefore gives $(P_n - P_\mu)(\hat{\psi}_\nu - \psi_\nu) = o_p(n^{-1/2})$ by conditional Chebyshev's inequality. The decomposition

$$\widehat{V}_{\mathrm{DR}} - V = (P_n - P_\mu)\psi_\nu + (P_n - P_\mu)(\hat{\psi}_\nu - \psi_\nu) + P_\mu \hat{\psi}_\nu - V$$

proves (12).

The residual term in ψ_ν has conditional mean zero given X, so it is uncorrelated with $m_\nu(X)$. Its conditional second moment is

$$\mathbb{E}_{P_\mu}\left[\frac{\nu(a \mid X)^2}{\mu(a \mid X)^2}\{r - r^*(X,a)\}^2 \middle| X\right] = \sum_a \frac{\nu(a \mid X)^2}{\mu(a \mid X)} \operatorname{Var}_D(r \mid X, a).$$

The law of total variance yields (13), and the i.i.d. central limit theorem proves the normal limit. Both scores are uniformly bounded by $M + 2M/c$. Their $L_2(P_\mu)$ difference is $o_p(1)$, so conditional laws of large numbers give $P_n\hat{\psi}_\nu^2 - P_n\psi_\nu^2 = o_p(1)$ and $P_n\hat{\psi}_\nu \to_p V$. Hence $\hat{\sigma}^2_{\mathrm{DR}} \to_p \sigma^2_{\mathrm{DR}}$, and Slutsky's theorem gives the studentized limit when the variance is positive. □

Proof of Proposition 3. Conditional on $\mathcal{C}_n$, write $\varepsilon_k = r_k - r^*(X_k, a_k)$ and $\hat{m}_\nu(X_k) = \sum_a \nu(a \mid X_k)\hat{r}(X_k, a)$. The fitted score is

$$\hat{\psi}_\nu(X_k, a_k, r_k) = \hat{m}_\nu(X_k) - w_k(a_k)\Delta(X_k, a_k) + w_k(a_k)\varepsilon_k.$$

The last term has mean zero conditional on X_k, a_k, giving the bias identity in (14). Cauchy–Schwarz under the empirical target-weighted measure, together with $\hat{\mu} \geq c$, gives its bound. By the law of total variance, the conditional variance of the score equals

$$\sum_a \mu(a \mid X_k) w_k(a)^2 \operatorname{Var}_D(r \mid X_k, a) + \operatorname{Var}_{a \sim \mu(\cdot \mid X_k)}\{w_k(a)\Delta(X_k, a)\}.$$

Conditional independence of the evaluation pairs gives $\operatorname{Var}(\widehat{V}_{\mathrm{DR}} \mid \mathcal{C}_n) = \sigma_n^2/n$. Adding the squared conditional bias proves (15). □

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